

# Vela IF820 Cable Replacement with SPP

## IF820 (EZ-Serial - SPP)

Quick Start Guide

v2.0

### 1 Introduction

EZ-Serial firmware platform supports BT Classic SPP service as a simple method to send and receive serial data over a Bluetooth connection. This operational mode is separated from the normal command mode where the API protocol may be used. When SPP Data Mode is active (aka Transparent Mode), any data received from an external host will be transmitted to the remote peer, and any data received from the remote peer will be sent out through the hardware serial interface to the external host.

### 2 Overview

This Quick Start Guide will walk users through connecting SPP Service from the IF820 module or dongle to an Android device, or a computer running a Windows 7 or later operating system. This is not meant to cover advanced connections with authentication and encryption enabled. For authentication and encryption configurations please reference the [User Guide - EZ Serial - Vela IF820 Series](#)

### 3 Requirements

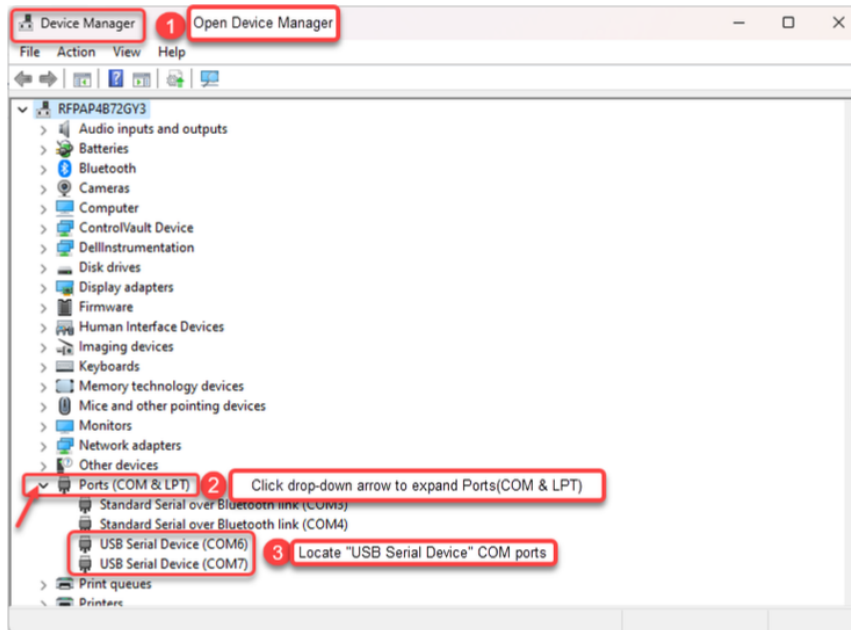
- One Vela IF820 Development Kit (DVK) or Vela IF820 Dongle. (A Vela IF820 module, integrated into a custom board, with EZ-Serial pins brought out, running EZ-Serial FW can also be used. See [Datasheet - Vela IF820 Series](#) for the required pins.)
- PC running Windows 7 or later – (for our example we will be using Windows 11)
- Android device – with **Bluetooth Serial Terminal** application installed (or similar application)
- [UwTerminalX](#) (v1.17 or later), or other Terminal Software
- [User Guide - EZ Serial - Vela IF820 Series](#)

## 4 Preparation

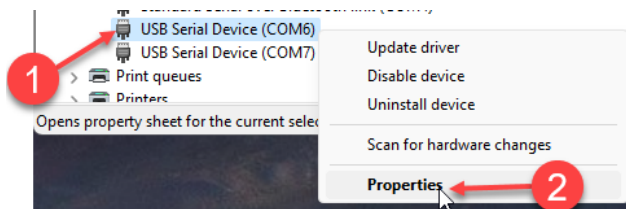
This application note assumes the Vela IF820 has the EZ-Serial firmware preloaded and updated to the most recent version – check the [EZ-Serial GitHub repository](#) for the most current release.

1. Verify the DVK jumpers are all in the default positions as per the [DVK-User Guide](#).
2. Connect the IF820 DVK to a PC using the supplied USB cable or insert the IF820 dongle into any available USB port.
3. Determine which COM port is UART1 (P\_UART)

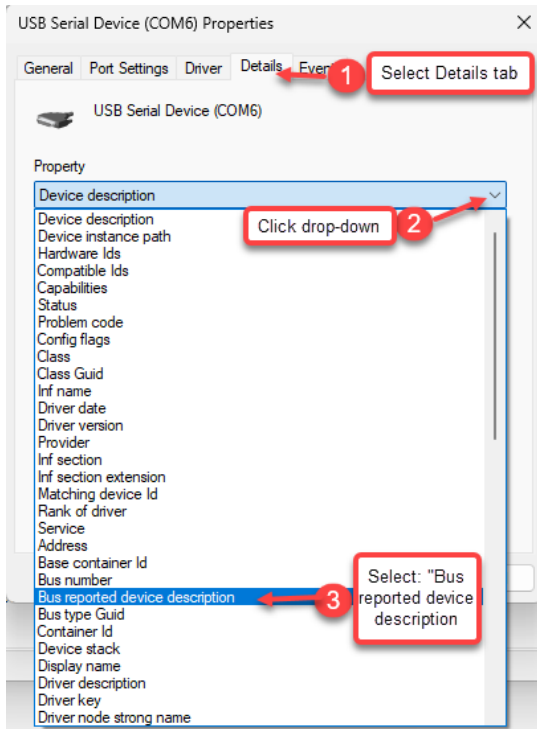
When the IF820 DVK is connected to a computer two COM ports are enumerated, UART0 which is the HCI UART, and UART1 also known as the "PUART". UART1/PUART is the COM port used to communicate with the module using EZ-Serial commands. To determine which COM port is UART1/PUART in Windows, open Windows Device Manager and locate the USB Serial Devices under the Ports (COM & LPT) section. (For Linux or MAC users reference the *IF820 DVK setup* section in the [Vela IF820 Software User Guide](#) to identify the COM ports.



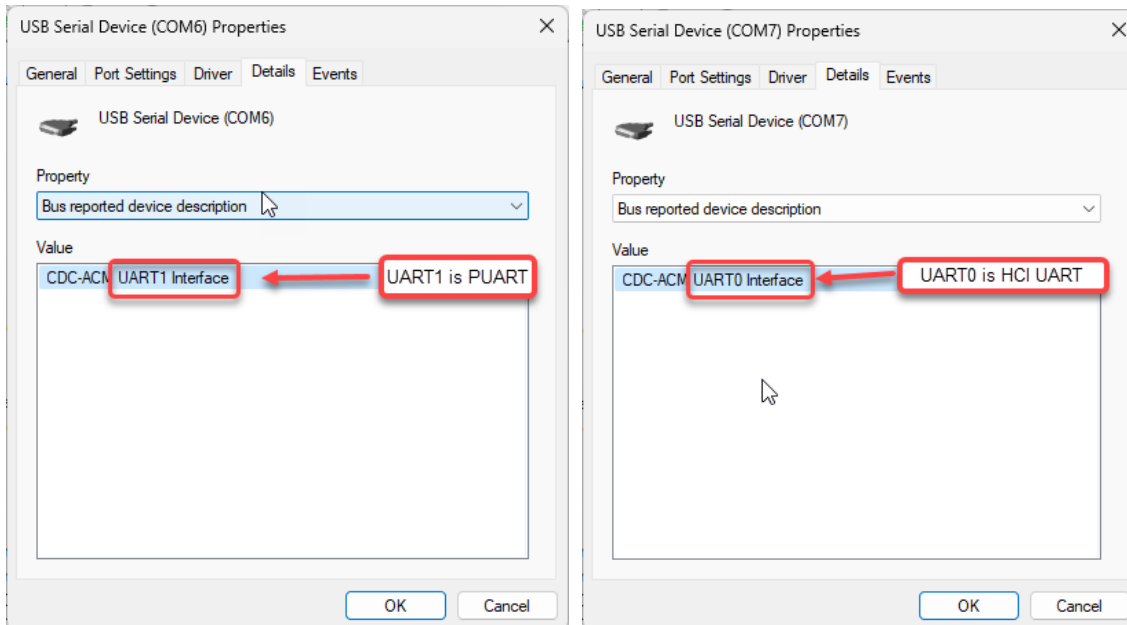
4. Right-click on one of the two COM ports and select **Properties** from the pop-up menu.



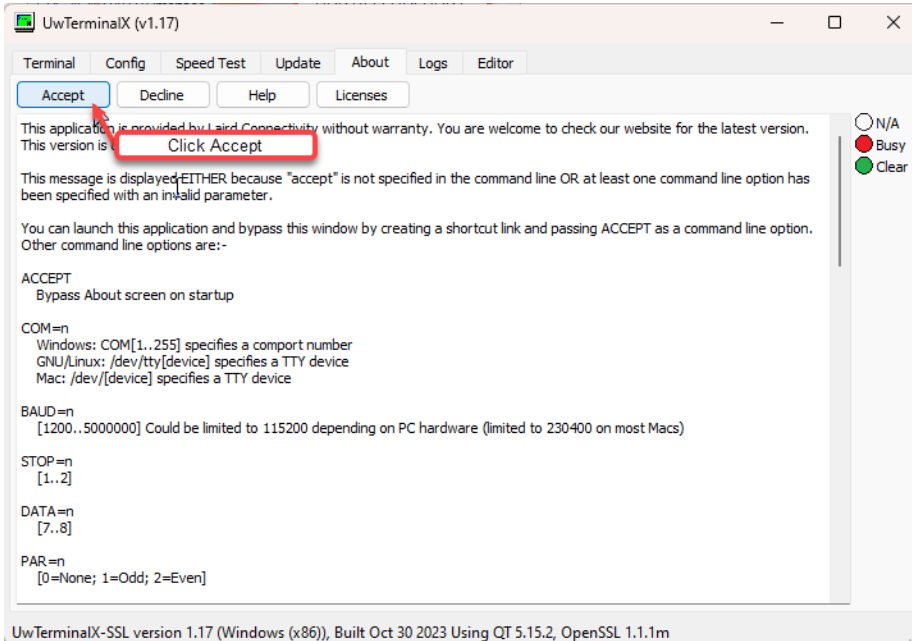
- Select the **Details** tab and expand the drop-down menu for **Property**, then select **Bus reported device description**.



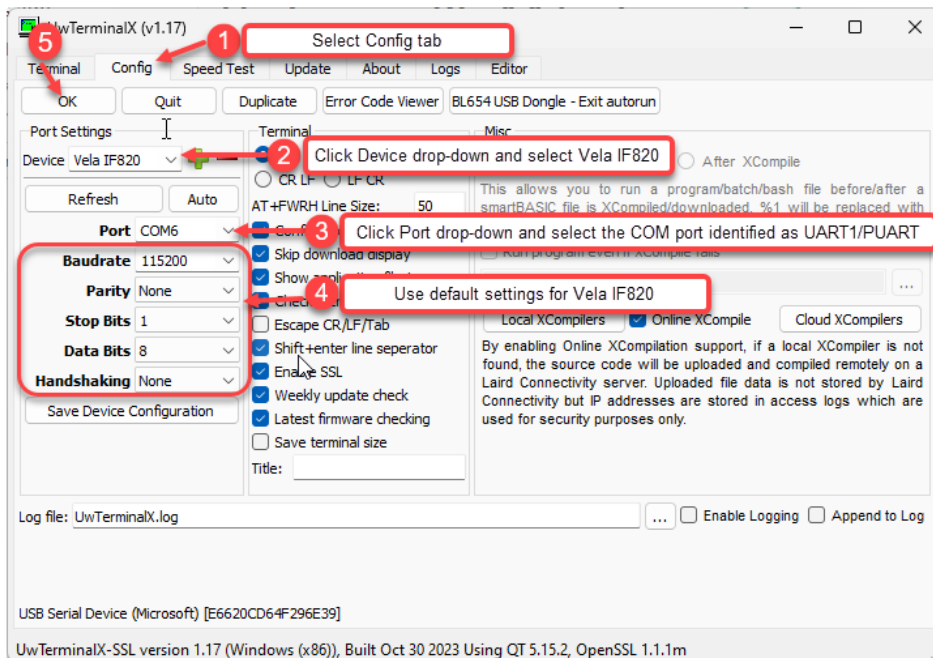
- If the Value contains "**UART1 Interface**", this is the PUART, which is used for EZ-Serial Command mode and Data mode – Note this COM port number as we will connect to it from our Terminal application. If it contains **UART0 Interface** this is the HCI UART, which is use for firmware flashing.



- Open **UwTerminalX** (or alternative serial terminal application such a Putty or RealTerm - All examples in this App Note are using UwTerminalX.) Click **Accept** on the home screen to accept the disclaimer.



- On the **Config** tab, click the drop-down for Devices and select **Vela IF820** to get default port settings for Vela IF820 modules, then click the drop-down **Device** under **Port Settings** and select the COM port identified as UART1/PUART in step 6 above. Use the default settings for Vela IF820. Click **OK** to open the port.



## 5 Verify Communication & EZ-Serial Firmware version

Send the following commands to test communication, disable EZ-Serial echo mode, and verify the firmware version (NOTE: commands are not case sensitive, can be in upper or lower case). The first two commands will echo **twice** in UwTerminalX because both EZ-Serial echo mode and UwTerminalX LocalEcho are enabled. Once EZ-Serial echo mode is disabled this will stop.

```
//PING //ping the module
SPEM,M=0 //disable EZ-Serial echo mode
/RBT (or press reset button on DVK) //Reboot the module to see boot message
```

The Vela IF820 modules come pre-loaded with Infineon's EZ-Serial firmware. Verify the firmware is the most current version available by reviewing the boot message: The value for "F" will contain a serial value declaring the firmware version loaded to the module. In the example below the version is 1.4.12.12, which was the most current version at the time this document was written. We always recommend working with the most current firmware version, which can be downloaded from the [Vela IF820 Firmware GitHub repository](#). If a newer version is available, please reference the [EZ-Serial Firmware Update](#) online guide to upgrade the firmware prior to starting this application note. See the [User Guide - EZ Serial - Vela IF820 Series](#) for full description of @R (response) and @E (event) outputs.

```

@E,000E,ASC,S=04,R=03
/ping
/ping
@R,001D,/PING,0000,R=00000022,F=8535
spem,m=0
spem,m=0
@R,000A,SPEM,00
/rbt
@R,000A,/RBT,0000
@E,0076,BOOT,E=01040C0C,S=03010000,P=0104,H=F2,C=00,A=D531AB62EC65,F=EZ-Serial-
VELA IF820 EXT V1.4.12.12 Sep 11 2023 10:21:21
@E,000E,ASC,S=03,R=03
@E,000E,ASC,S=04,R=03
@E,000E,ASC,S=00,R=03
  
```

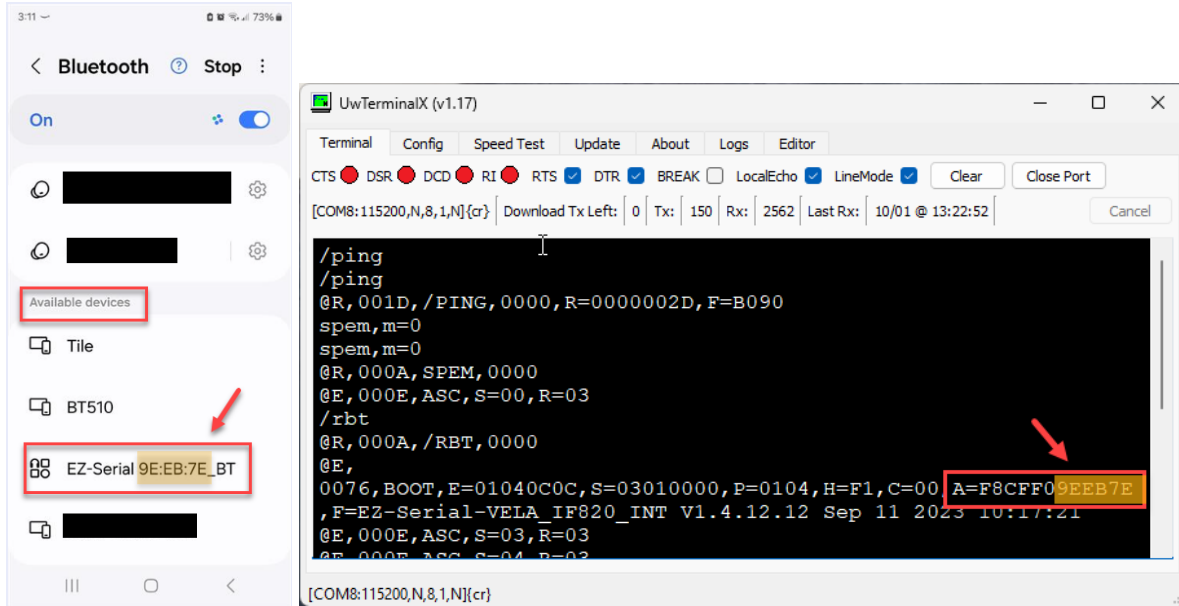
## 6 Connect & Send Data via SPP Connection with Android device

By default, EZ-Serial advertises for an SPP connection. To connect with an Android device, follow the steps below:

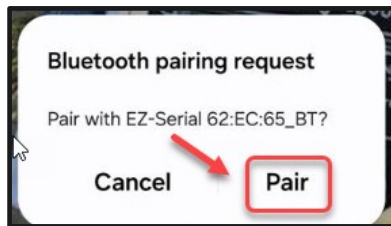
1. Open Bluetooth settings to pair with the EZ-Serial device. Ensure Bluetooth® is enabled on the Android device.



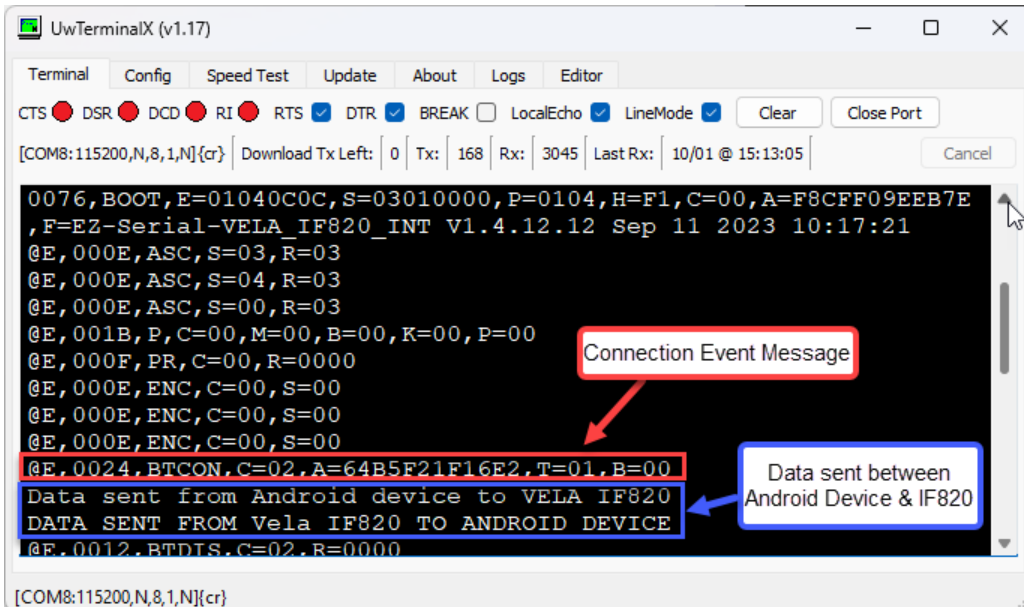
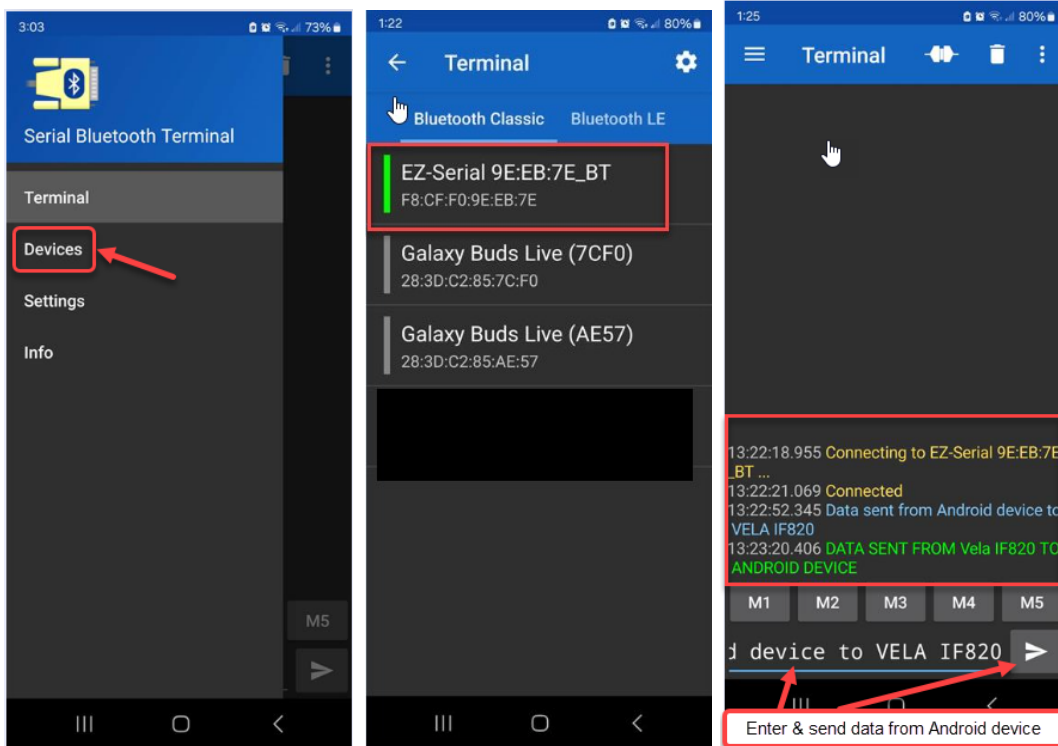
2. With Bluetooth enabled, locate **EZ-Serial xx:xx:xx\_BT** in the list of *Available Devices* under Bluetooth Settings. Click on **EZ-Serial xx:xx:xx\_BT**, to pair with the EZ-Serial device. NOTE: xx:xx:xx is the last three bytes of the IF820 Bluetooth Address in the Boot message.



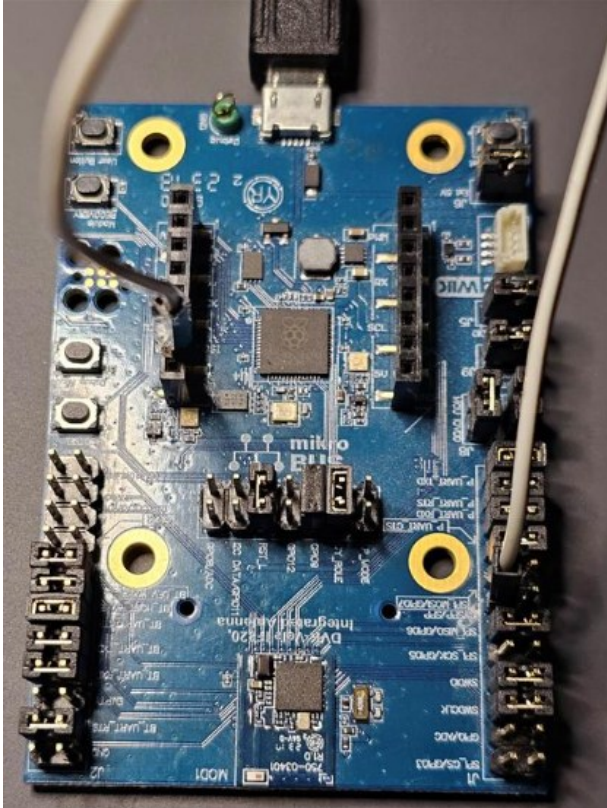
3. If a Bluetooth pairing request message is received on the Android device select the appropriate response to allow it to pair.



- After pairing, open a Serial Terminal Application, such as **Serial Bluetooth Terminal**, to make an SPP connection between the Android device and the IF820. Once the device is selected the Terminal window will open showing the devices have connected and now send and receive data over the SPP Data link, as shown in the images below.

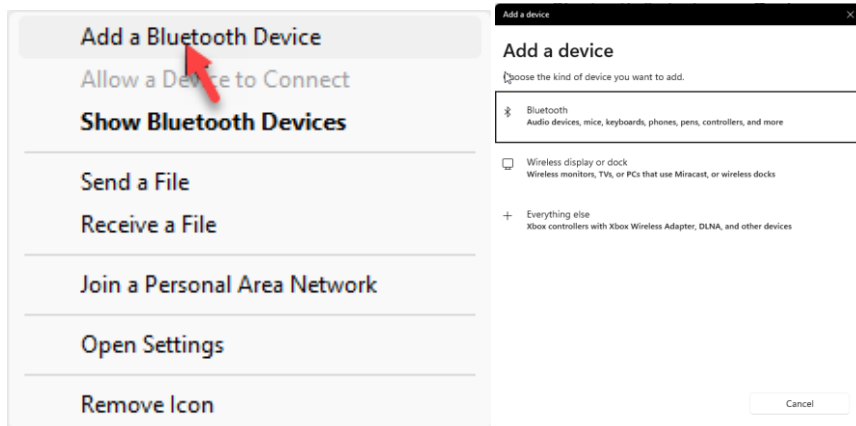


- To disconnect, either select disconnect in the Android application, or remove the CYSPP/SPP jumper on the IF820-DVK, to isolate the signal, and toggle the CYSPP/SPP of the IF820 pin low-to-high, as shown in the image below This will cause the module to disconnect and return to Command Mode (not available on the IF820 Dongle).

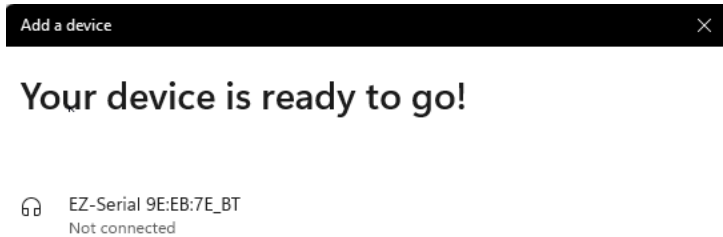


## 7 Connect & Send Data via SPP Connection with Windows PC (Windows 11)

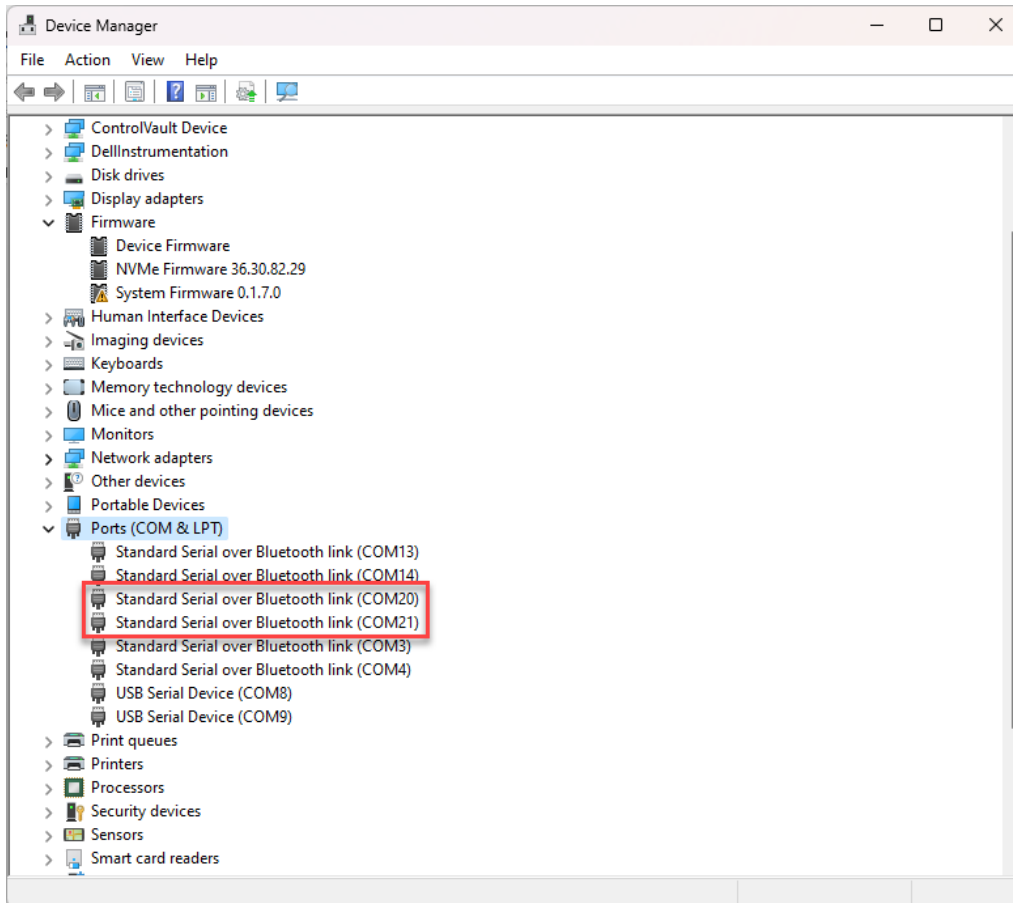
- Open Bluetooth settings on the Windows PC and select **Add a Bluetooth Device**. Follow the prompts until it lets there is an option to select the EZ-Serial xx:xx:xx\_BT device with the BT\_ADDR of the IF820 module from the Boot message.



- When the device is ready to open, Windows displays the following prompt:



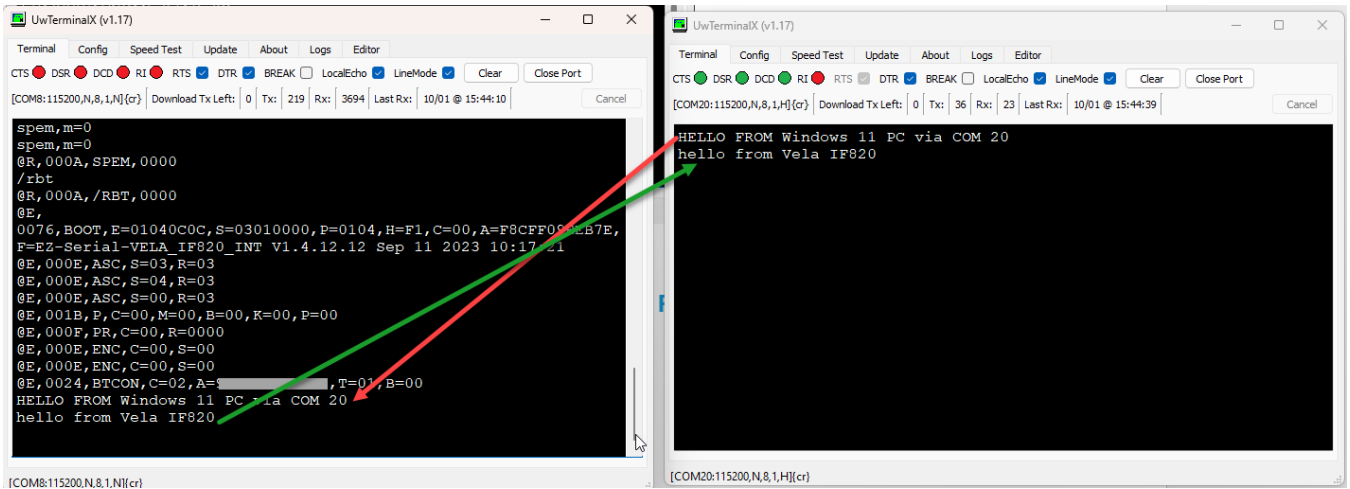
- Open **Windows Device Manager** and expand the COM ports to determine the COM ports assigned to the SPP connection – there should be two **Standard Serial over Bluetooth link** ports created when the SPP connection is created. Usually, the lower numbered COM port is the one that should be connected to, using a terminal application such as UwTerminalX.



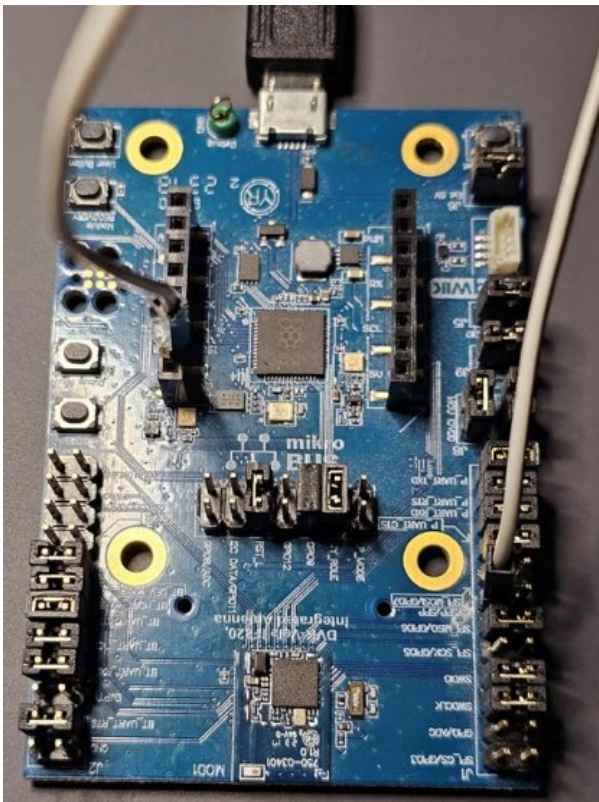
- Open a terminal application and select the COM port determined in step 2 above. An event message will be displayed in the IF820 terminal if the correct COM port has been selected, showing a connection has occurred, if not try the other COM port.

```
@E, 0024, BTCON, C=02, A=xxxxxxxxxxxx, T=01, B=00
```

- Once the BTCON event is received the module is in Data mode and data can now be exchanged over the Bluetooth connection.



- To disconnect, either close the COM port on the PC side of the connection or remove the CYSPP/SPP jumper on the IF820-DVK, to isolate the signal, and toggle the CYSPP/SPP of the IF820 pin low-to-high, as shown in the image below This will cause the module to disconnect and return to Command Mode (not available on the IF820 Dongle).



## 8 Revision History

| Version | Date        | Notes             | Contributors   | Approver       |
|---------|-------------|-------------------|----------------|----------------|
| 1.0     | 18 Jan 2023 | Initial Release   | Rikki Horrigan | Jonathan Kaye  |
| 2.0     | 3 Mar 2025  | Ezurio rebranding | Dave Drogowski | Dave Drogowski |